

TONY HANNA

Biomedical Engineer

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📍 Sydney, NSW, 2000

CAREER OBJECTIVE

Biomedical Engineering graduate from the University of Technology Sydney with hands-on experience across both industry and laboratory environments, including medical device servicing in clinical settings and research within cleanroom, vacuum and PC2 laboratories. Seeking to apply my technical knowledge, practical engineering experience and problem-solving skills while growing within a multidisciplinary engineering or healthcare organisation.

EDUCATION

University of Technology Sydney | 2023-2026

Bachelor of Engineering (Honours) - Biomedical Engineering

Cumulative GPA 6.11/7.00 | WAM 80.0/100 | Dean's List 2026

WORK EXPERIENCE

Institute of Biomedical Materials and Devices | June 2026 - September 2026

Dayong Jin Student Fellowship - R&D Engineer

- Gained hands-on experience in microdevice fabrication using photolithography and other microfabrication techniques within laboratory and cleanroom environments.
- Worked with micro-scale components and conductive features, developing practical skills in fabrication, handling, testing and troubleshooting.
- Applied engineering principles across electrical, mechanical and materials-focused aspects of device development.
- Built experience in experimental testing, process refinement, technical problem-solving and working with specialised laboratory equipment.

Advantech Engineering | January 2026 - April 2026

Biomedical Engineering Intern

- Serviced, tested and repaired medical devices across multiple healthcare clinics, diagnosing faults and restoring equipment to safe working condition.
- Worked directly with clinic staff and customers to understand equipment issues, communicate findings and minimise downtime.
- Applied manufacturer procedures and structured troubleshooting methods when assessing and repairing biomedical equipment.
- Maintained accurate service documentation using Zoho CRM while managing work across multiple customer sites.

Smithfield Autoglass | January 2022 - December 2025

Autoglass Technician

- Worked efficiently with hand and power tools to safely remove, prepare and replace vehicle windscreens while maintaining attention to quality and workmanship.
- Provided professional customer service to a broad range of clients and demographics, communicating clearly and ensuring a positive customer experience throughout the service process.

ACADEMIC PROJECTS

Bone Scaffold Integration (Capstone Project)

- Developing a biomimetic bone scaffold using hydrogels, peptides and other biomaterials, supported by literature review and experimental research.
- Applying CAD knowledge alongside bioprinting, biomaterial preparation and mechanical testing in PC2 and sterile laboratory environments.

Machine Learning for Anaemia Detection

- Developed an image-processing and machine-learning workflow for non-invasive anaemia detection using conjunctival images.
- Applied image segmentation, feature analysis and multiple machine-learning models while evaluating performance and dataset quality.

ECG Monitoring & Signal Acquisition System

- Developed a functional ECG monitoring system combining biomedical instrumentation, electrical circuitry and a LabVIEW-based interface.
- Integrated and troubleshoot hardware and software components to acquire, process and display physiological signals in real time.

Bioinformatics Case Study – Genomic Diagnosis of Leukaemia

- Analysed genomic and clinical data from a leukaemia case study, using sequencing and variant analysis to support diagnosis and investigate disease-associated mutations.
- Applied bioinformatics workflows including sequence alignment, variant calling and genomic interpretation using tools such as Galaxy, Bowtie2, Mutect2 and public genomic databases.

Nominated Projects at UTS Tech Festival 2025 and 2026

- **2025:** Developed and presented 3D bioprinted brain models designed to investigate CTE and the effects of repeated traumatic impact on neural tissue.
- **2026:** Developed and presented a lateral flow assay concept for rapid concussion assessment, exploring the potential use of biomarkers to support post-impact screening in athletes.

SKILLS

Engineering & Technical: Troubleshooting, fault finding, equipment testing, technical servicing, CAD, fabrication, prototyping and mechanical testing.

Laboratory & Research: Experimental testing, biomaterial preparation, photolithography, cleanroom and PC2 laboratory experience, literature review and technical documentation.

Programming & Data: Python, Java, SQL, MATLAB, machine learning, bioinformatics, data analysis and Linux environments.

Software: SolidWorks, Fusion 360, AutoCAD, LabVIEW, ImageJ and Microsoft Office.

Professional: Problem-solving, attention to detail, communication, customer service, teamwork, adaptability, time management and independent work.

REFERENCES

Available upon request.